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Pharmacotherapy and overdose risk following new opioid use disorder diagnosis among Medicaid beneficiaries

Peter Treitler^a, Hillary Samples^{b,c}, Richard Hermida^b, Jennifer Miles^{b,d}, Elizabeth M. Stone^{b,e}, Stephen Crystal^{b,c,f}

^aDepartment of Human Behavior, Research, and Policy, Boston University School of Social Work, Boston, MA, USA

^bInstitute for Health, Health Care Policy and Aging Research, Rutgers University, New Brunswick, NJ, USA

^cSchool of Public Health, Rutgers University, New Brunswick, NJ, USA

^dDepartment of Family Medicine and Community Health, Rutgers Robert Wood Johnson Medical School, New Brunswick, NJ, USA

^eDepartment of Psychiatry, Rutgers Robert Wood Johnson Medical School, New Brunswick, NJ, USA

^fSchool of Social Work, Rutgers University, New Brunswick, NJ, USA

Abstract

Background: Timely initiation and continuation of medications for opioid use disorder (MOUD) following a new opioid use disorder (OUD) diagnosis reduces overdose risk.

Objectives: To examine patient- and community-level factors associated with timely MOUD receipt and the association of these factors and MOUD modality (buprenorphine, methadone, naltrexone) with subsequent overdose.

Methods: This cohort study included Medicaid enrollees in 44 states aged 18–64 with a new OUD diagnosis between April 2016 and December 2019. Multivariable Cox Proportional Hazards models assessed time (in days) to MOUD receipt and medically-treated nonfatal overdose stratified by involvement of heroin/synthetic vs. other (primarily prescription) opioids only. Models adjusted for patient, county, and state-level covariates, and models examining overdose additionally adjusted for time-varying (daily) MOUD receipt.

Results: Of 1,172,200 Medicaid enrollees with a new OUD diagnosis, 51.8% were female, 55.8% White, and 42.2% aged 30–44. Most (69.2%) did not receive MOUD within 180 days. MOUD receipt was lower among Black (Hazard Ratio [HR] = 0.68, 95% Confidence Interval [CI] 0.67–0.69) and Hispanic (HR = 0.91, 95%CI = 0.90–0.92) compared to White enrollees. During follow-up, 3.2% of enrollees experienced heroin/synthetic opioid overdose, with reduced risk more strongly associated with methadone (HR = 0.14, 95%CI = 0.12–0.15) and buprenorphine (HR = 0.23, 95%CI = 0.22–0.25) than naltrexone (HR = 0.70, 95%CI 0.64–0.77).

Conclusion: Use of methadone and buprenorphine, and to a lesser extent naltrexone, are associated with lower overdose risk among Medicaid beneficiaries newly diagnosed with OUD. State and local policies aimed at increasing timely, equitable, and sustained treatment receipt are needed to address the opioid epidemic and reduce disparities.

Keywords

Opioid use disorder; medications for opioid use disorder; MOUD; Medicaid; Overdose

Introduction

Despite recent improvements, opioid overdose fatalities remain historically high (1). Over 80,000 fatal opioid overdoses occurred in the US in 2023, with an increasing proportion occurring among racial and ethnic minorities (2, 3). Clinical trials have consistently found medications for opioid use disorder (MOUD) to be highly effective at reducing overdose risk (4, 5). However, additional research is needed to understand MOUD treatment outcomes in large usual-care populations, including whether protective effects differ across the three medications (methadone, buprenorphine, naltrexone).

MOUD remains underutilized despite substantial policy and provider efforts toward increasing access, with disparities in MOUD use by individual (e.g., race/ethnicity, clinical conditions) and structural factors (e.g., geography) (6). Uneven access has been observed based on comorbid psychiatric and substance use disorders (SUD), overdose history, chronic pain, and hepatitis C (7–9). Additionally, MOUD use is lower among Black compared to White patients, who have greater geographic access to buprenorphine (10–14). Other factors may contribute to geographic variation observed at the state level (12, 15–17).

In turn, overdose risk is influenced by limited treatment availability (e.g., buprenorphine prescribers, opioid treatment programs) (18), delays in accessing treatment (6, 19), and a similarly complex mix of individual and structural factors, including community-level social disadvantage (20, 21), fentanyl in the illicit drug supply (22), and patient-level factors such as injection drug use, higher-frequency acute care use, comorbid psychiatric disorders and SUD, and concurrent use of opioids, benzodiazepines, and other drugs causing respiratory depression (8, 19, 23–25).

While much research has examined MOUD uptake, treatment outcomes, and overdose risk, few studies have examined MOUD use in national, usual care populations of individuals with OUD. Specifically, understanding these issues in Medicaid is critical given the social and health challenges faced by this population who account for nearly 4 in 10 US individuals with OUD (26). One recent analysis found wide state and county-level variation in any MOUD receipt among Medicaid beneficiaries with OUD (15). However, this analysis measured MOUD receipt at any point in a calendar year when an OUD diagnosis was observed rather than MOUD receipt following a new diagnosis. The effectiveness of the various MOUD modalities in preventing overdose during active treatment is not fully understood, with some studies showing greater risk reduction with buprenorphine compared to naltrexone (27) and others showing similar risk for both medications (28–30). No known studies have compared all three medications in the national Medicaid population. This study

seeks to address these gaps by examining patient and community factors associated with timely MOUD receipt and opioid overdose following new OUD diagnosis, in national U.S. Medicaid data from 2016–2019. By including time-varying measures of MOUD receipt in a time-to-event framework, this study compares associations of the three MOUD modalities with overdose.

Materials and methods

This study was reviewed and approved by the Rutgers University Institutional Review Board and follows the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines for cohort studies. The requirement for informed consent was waived due to the use of deidentified administrative data.

Data and study cohort

Study data included national Transformed Medicaid Statistical Information System (T-MSIS) Analytic Files (TAF; Release 2) from November 2015 to December 2019. Data included information on enrollee characteristics, diagnoses, and service and prescription drug use, spanning the time period from the start of Medicaid expansion to just before the healthcare disruptions of the COVID-19 pandemic. We excluded state-years with missing Medicare enrollment information precluding the identification of dually eligible enrollees (Alabama: 2016–2019, Arkansas: 2016; Michigan: 2016; Utah: 2016–2019), missing methadone claims (Illinois: 2016–2019; New York: 2016–2019), missing county (Rhode Island: 2016–2019, Vermont: 2016–2019, Wyoming: 2016–2019), or patient identifier changes precluding cross-year claims linkages between 2015 and later years (Louisiana and Mississippi: April–June 2016) (Table A1).

The study cohort included Medicaid enrollees aged 18–64 with a new (index) diagnosis of OUD between April 1, 2016 to December 31, 2019. To capture a broad cohort of individuals with likely “new” OUD, we defined new diagnosis as the first diagnosis occurring after 180+ days without an OUD diagnosis or MOUD receipt. Enrollees also had to be continuously enrolled in Medicaid for the month of the new OUD diagnosis and 6 prior months and have no dual Medicare enrollment throughout the lookback and follow-up periods. Individuals aged 65 and older and those dually enrolled in Medicare were excluded because Medicare serves as the primary payer for these enrollees, and Medicare claims were not available for analysis.

Measures

Patient-level variables included the year of the index OUD diagnosis and demographics (race/ethnicity [White, Black, Hispanic, Other or Unknown], age group [12–17, 18–25, 26–44, 45–54, 55–64], and sex [male, female]). In the 180-day baseline period before new OUD diagnosis, we assessed the presence of comorbid SUD (alcohol; cannabis; cocaine; other stimulants; and sedative, hypnotic or anxiolytic), psychiatric disorders (anxiety, autism, bipolar disorder, intellectual disability, depression, personality disorder, schizophrenia, and any other psychiatric disorder), and chronic medical conditions (asthma, cerebrovascular disorders, chronic pain, chronic obstructive pulmonary disease, diabetes,

heart failure, hepatitis C, HIV, hypertension, pneumonia, and sleep apnea). We further assessed baseline history of medically-treated opioid overdose (any heroin/synthetic opioid involvement, other opioids only) and non-opioid drug overdose, and any receipt of prescription antidepressants, antipsychotics, benzodiazepines, or opioids (excluding buprenorphine medications specifically indicated for OUD). Diagnoses and service codes used to identify patient characteristics are shown in Tables A2–A3.

State-level variables included fentanyl penetration (measured annually as the ratio of fentanyl-to-heroin law enforcement seizures, reported to the National Forensic Laboratory Information System) (31), Medicaid expansion status (indexed to the date of the new OUD diagnosis) (32), and region (Northeast, Midwest, South, and West). County-level variables included median age, percentage non-Hispanic White, social deprivation index (SDI) percentile (a composite measure of socioeconomic disadvantage, with higher scores indicating greater disadvantage (Appendix A, Supplementary Methods) (33), and population density using rural-urban continuum codes (metropolitan, non-metropolitan) (34). County demographics and SDI were calculated Area Health Resources File (AHRF) data from corresponding years (35). Rural-urban continuum codes were derived from the 2013 Department of Agriculture's Economic Research Service, also contained in the AHRF.

Study outcomes assessed in the 180-day follow-up period after new OUD diagnosis were time (in days) to MOUD receipt and medically-treated nonfatal opioid overdose. We stratified the overdose outcome by involvement of any heroin/synthetic opioids vs. other opioids only, to differentiate between overdoses involving illicitly-manufactured opioids and those involving prescription formulations only. Factors associated with overdose and the appropriate intervention may differ for these categories. MOUD included buprenorphine (excluding formulations primarily indicated for pain), methadone dispensed in opioid treatment programs, and extended release (ER) naltrexone. MOUD receipt was assessed using medication calendars with daily indicators of available supply over the follow-up period, determined using prescription dispensing or administration date and days' supply variables (Appendix A, Supplementary Methods). Service codes and drug names used in defining these outcomes are shown in Tables A2–A3. Additionally, we measured daily supply of prescription opioids in the follow-up period to control for the influence on receipt of MOUD and risk of overdose.

Statistical analysis

Descriptive statistics were calculated for the full cohort and subgroups with each outcome of interest. Cox proportional hazards models estimated time to MOUD receipt and opioid overdose within 180 days of new OUD diagnosis. Models assessing MOUD receipt adjusted for all patient, county, and state-level characteristics described above, including the time-varying measure of prescription opioid receipt in the follow-up period. Models examining medically-treated overdose additionally included time-varying variables representing daily supply of each MOUD during the 180-day follow-up period, with a separate indicator for each day (1 = MOUD supply, 0 = no supply). These variables estimate the association between MOUD exposure and overdose risk on each day of follow-up, with individuals classified as unexposed on days without MOUD supply. This approach avoids immortal time

bias by ensuring that person-time without MOUD is correctly attributed to the unexposed group (36, 37). Follow-up was right-censored at 180 days, Medicaid disenrollment, the end of the study period, or occurrence of the outcome. Standard errors were clustered on county of residence. While we adjusted for a large set of variables potentially associated with MOUD receipt and overdose risk, select variables are shown in the manuscript tables for brevity; see Tables A4–A5 for complete tables.

We conducted four sensitivity analyses. First, we allowed individuals to contribute more than 1 “new” OUD episode to the dataset. Patients could contribute 1 episode per year and the earliest was taken if there was more than 1. Other inclusion/exclusion criteria still applied, including the requirement for no OUD diagnosis in the 180 days prior to the “new” OUD diagnosis. (Table A6). Second, we excluded additional state-years with data quality concerns identified in the TAF Data Quality (DQ) Atlas (38) and internal assessments (39). This analysis included data from 24 states (including DC) in all 4 study years and 40 states in at least 1 year (Table A1, A7). Third, because many patients who experience opioid overdose have OUD, we conducted a sensitivity analysis excluding patients with opioid overdose in the baseline period even if they did not have an OUD diagnosis or receive MOUD (Table A8). Fourth, due to known challenges in estimating methadone days’ supply from claims data (40), we tested an alternate definition in which each methadone dispensing claim was assumed to represent a single day of supply (Table A9). Analyses used a 2-tailed significance threshold of $p < .05$ and were performed from January 2023 to July 2025 using SAS Enterprise Guide Version 8.3.

Results

Between April 1, 2016–December 31, 2019, 1,172,200 Medicaid enrollees had a new OUD diagnosis, of whom 361,034 (30.8%) received MOUD 37,641 (3.2%) experienced a heroin/synthetic opioid overdose, and 21,956 (1.9%) experienced an overdose involving other opioids within 180 days (Table 1). Among those who received MOUD, 216,095 (59.9%) initiated treatment within 0–7 days of their OUD diagnosis 65,880 (18.2%) within 8–30 days 45,931 (12.7%) within 31–90 days, and 33,128 (9.2%) within 91–180 days. The cohort was 51.8% female, 14.3% Black, 8.5% Hispanic, 55.8% White, and the plurality were ages 30–44 (42.2%). Common comorbidities included chronic pain (49.2%), anxiety (22.5%), and depression (17.6%); 40.0% and 18.2% had baseline opioid and benzodiazepine prescriptions, respectively. Counties of residence had a median of 71.3% White residents, and an SDI percentile of 54.

MOUD receipt

MOUD receipt increased annually (2019 vs. 2016 HR = 1.21, 95%CI 1.20–1.22) but was less likely for Black (HR = 0.68, 95%CI 0.67–0.69), Hispanic (HR = 0.91, 95%CI 0.90–0.92), and other race (HR = 0.87, 95%CI 0.86–0.87) compared to White enrollees (Table 2). Compared to those aged 55–64, enrollees aged 18–29 (HR = 1.06, 95%CI 1.05–1.07) and 30–44 (HR = 1.24, 95%CI 1.23–1.25) were more likely to receive MOUD. Individuals with comorbid SUDs or psychiatric disorders were less likely to receive MOUD.

However, MOUD receipt was higher for individuals with a history of medically-treated heroin/synthetic opioid overdose (HR = 1.06, 95%CI 1.03–1.08).

Patients were more likely to receive MOUD if they resided in counties with higher percentages of White residents (HR = 1.03 per 10% points, 95%CI 1.03–1.04) and higher SDI score (HR = 1.01 per 10% points, 95%CI 1.01–1.01). Patients residing in states with a greater fentanyl-to-heroin seizure ratio (HR = 1.11, 95%CI 1.11–1.12) and those that expanded Medicaid (HR = 1.24, 95%CI 1.23–1.25) were more likely to receive MOUD. Patients had a lower likelihood of MOUD receipt if they resided outside the Northeast. Sensitivity analyses had similar results as the main analysis (Tables A6–A9).

Medically-treated overdoses involving heroin/synthetic opioids

Risk of heroin/synthetic opioid overdose within 180 days of new OUD diagnosis was significantly reduced while receiving MOUD. Hazard of overdose was 86% lower while receiving methadone (HR = 0.14, 95%CI 0.12–0.15), 77% lower for buprenorphine (HR = 0.23, 95%CI 0.22–0.25), and 30% lower for naltrexone (HR = 0.70, 95%CI 0.64–0.77) (Table 2, Figure 1). Hazard of heroin/synthetic overdose decreased between 2016 and 2019 (HR = 0.71, 95%CI 0.68–0.73), coinciding with the increase in MOUD receipt during the same period. Risk was lower for Black (HR = 0.68, 95%CI 0.66–0.70), Hispanic (HR = 0.89, 95%CI 0.85–0.92), and other race (HR = 0.94, 95%CI 0.92–0.97) compared to White enrollees, as well as for female compared to male enrollees (HR = 0.65, 95%CI 0.64–0.66). Beneficiaries with opioid prescriptions in the baseline (HR = 0.68, 95%CI 0.66–0.70) and follow-up (HR = 0.56, 95%CI 0.53–0.59) periods had lower hazards of heroin/synthetic opioid overdose. Risk was especially high for individuals with a prior (baseline) heroin/synthetic opioid overdose (HR = 4.85, 95%CI 4.69–5.02). Hazard of heroin/synthetic opioid overdose was lower for patients living in counties with more White residents (HR = 0.93 per 10% points, 95%CI 0.92–0.94) and higher SDI percentile (HR = 0.97 per 10% points, 95%CI 0.97–0.98), and higher for those in states that expanded Medicaid (HR = 1.58, 95% CI 1.52–1.64). Results from sensitivity analyses were substantively similar to the main analyses (Tables A6–A9).

Medically-treated overdoses involving other opioids

Characteristics associated with other opioid overdoses were generally similar to those associated with heroin/synthetic opioids, with a few notable exceptions. Anxiety diagnosis (HR = 1.05, 95%CI 1.01–1.08), baseline benzodiazepine prescriptions (HR = 1.32, 95%CI 1.28–1.37), and opioid prescriptions during follow-up (HR = 1.39, 95%CI 1.34–1.45) were associated with higher overdose risk. Overdose risk was lower on days with MOUD supply than those without supply, but the magnitude of the association for methadone (HR = 0.39) was less than for heroin/synthetic overdoses (HR = 0.14). At the community level, individuals in states with a greater fentanyl-to-heroin seizure ratio had higher risk of overdose (HR = 1.05, 95%CI 1.02–1.07), while overdose risk was lower in states that had expanded Medicaid (HR = 0.92, 95%CI 0.88–0.96).

Discussion

In the Medicaid population, the proportion of individuals with newly-diagnosed OUD receiving medication within 180 days increased from 27.4% to 33.6% during the study period. Despite this modest improvement, substantial disparities were observed by race/ethnicity, with lower likelihood of MOUD receipt for Black and Hispanic compared to White individuals, and for patients with comorbid SUD or psychiatric illness. The importance of addressing disparities in timely access to MOUD is underscored by the significant reduction in risk of overdose associated with receipt of any of the three medications.

In addition to differences by patient characteristics, we observed higher MOUD receipt in states with greater fentanyl penetration (e.g., states in the Northeast), and substantially lower MOUD receipt in the Western, Midwestern, and Southern regions of the US compared with the Northeast. Areas with high rates of fentanyl seizures often have high rates of opioid overdose mortality (22, 41). While prior evidence indicates the Northeast region of the US has high rates of treatment availability in areas with correspondingly high overdose rates, discrepancies between overdose rates and treatment availability in other regions remain a considerable public health concern (18). Residence in Medicaid expansion states was associated with higher treatment rates, underscoring the significance of Medicaid in improving treatment access. As the dominant payer of OUD services, Medicaid expansion can increase treatment access for all Medicaid enrollees (not just those newly enrolled) by increasing the supply of providers (42). States that expand Medicaid may also be more likely to implement other policies that increase OUD treatment access, such as 1115 waivers that expand service and coverage options (43).

The finding that each MOUD is associated with lower overdose risk supports results from earlier studies (4, 5, 19, 23). In this Medicaid population, the effect sizes for methadone and buprenorphine utilization are particularly noteworthy. These opioid agonist treatments were associated with 86% (methadone) and 77% (buprenorphine) reductions in risk of heroin/synthetic opioid overdose, far surpassing the 30% reduction for opioid antagonist treatment (naltrexone). While prior studies comparing naltrexone and opioid agonists have been somewhat mixed (27, 44), our results suggest greater effectiveness of opioid agonist MOUD treatment. Other studies have shown comparably protective effects on overdose for patients actively treated with buprenorphine or ER naltrexone (28–30). Further research is needed to understand why naltrexone is associated with a smaller reduction in overdose risk compared to other MOUD modalities. Case reviews of overdose deaths among patients on naltrexone suggest that risk may increase as serum levels decline toward the end of a dosing interval. Additionally, overdose risk may be heightened among patients with low opioid tolerance who consume large opioid doses, or in combination with sedatives, in an effort to override naltrexone's opioid receptor blockade (45, 46).

The large risk reduction associated with methadone is particularly striking in the context of recent policy and provider efforts to expand MOUD access that overwhelmingly focus on buprenorphine and naltrexone, in part due to fewer regulations and the ability to prescribe them in any setting. Although recent changes to opioid treatment program (OTP)

guidelines have reduced barriers to methadone (e.g., extended take-home doses) (47), most patients are still required to attend the clinic frequently, deterring many from initiating or continuing treatment. The U.S. is unique among comparable nations in restricting methadone delivery to highly regulated OTPs. In Canada, for example, methadone regulation resembles U.S. policies for prescribing and dispensing buprenorphine (48). If proposals to reform methadone regulations in the U.S (49). were implemented, access to lifesaving MOUD could be greatly expanded.

Alongside increases in MOUD from 2016–2019, we found decreases in medically-treated opioid overdose among those newly diagnosed with OUD. These findings contrast with national increases in overdose deaths during the same period (50). Reasons for this inconsistency are unclear, but it is possible that patients identified as having OUD were more rapidly linked to MOUD over time, thereby reducing their risk of overdose. This would suggest that even as overdose mortality continues to rise, the health system's effectiveness at identifying patients with OUD and initiating treatment may be improving, possibly leading to reductions in overdose mortality. Additionally, expanded access to naloxone (51) or increasing lethality of the drug supply during this period could have led to fewer people treated in acute care settings for nonfatal overdose if reversal was either successful or not possible.

Risk of overdose after new OUD diagnosis was lower among Black and Hispanic compared to White enrollees, yet overdose risk was lower for enrollees residing in counties with higher percentages of White residents. A few small studies have found that differential patterns of use in racial and ethnic groups may account for patient-level differences during the study period; for example, injection drug use or frequent and high-dose opioid use may be less likely among Black compared to White individuals (52, 53). At the county level, the association of higher proportion of White residents with lower overdose risk is likely explained by greater MOUD availability in those counties. It is important to note that racial and ethnic differences in overdose risk may have shifted since the study period as more-recent overdose data show that opioid overdose rates among Black individuals have exceeded those of White individuals since 2020 (54, 55).

Overdose risk was also lower for female compared to male enrollees, possibly a function of lower-risk drug use (56), lower-dose opioid prescribing (57), earlier diagnosis and treatment initiation, and better treatment adherence (58). Further research is needed to understand demographic differences in overdose risk and their implications for intervention design. Additional patient factors associated with higher overdose risk included prior opioid overdose and hepatitis C, both likely markers of OUD severity. Assessing history of OUD-related health consequences can help identify patients at greatest risk and in need of timely intervention.

Unexpectedly, we found no association between state fentanyl penetration and heroin/synthetic opioid overdose, despite finding a positive association with other opioid overdose. Providers may document synthetic opioid overdoses inconsistently (59) and it is possible that some were misclassified as other (semi-synthetic) opioid overdoses. Additionally, we relied on fentanyl seizure data to determine fentanyl penetration, which is subject to

measurement error due to variation in enforcement and testing practices across jurisdictions and over time (41, 60). Finally, despite greater receipt of MOUD among individuals residing in Medicaid expansion states, and the protective effect of MOUD on overdose, we found that heroin/synthetic (but not other opioid) overdose risk was higher for patients living in Medicaid expansion states. However, this likely reflects higher rates of fentanyl spread in regions of the country that expanded Medicaid, particularly in the northeast (60).

Limitations

While use of national Medicaid data provides an important population-based perspective, our estimates are subject to the typical limitations of claims data, including potential missed or misclassified OUD diagnoses. We relied on the Healthcare Cost and Utilization Project standard for identifying OUD in medical claims, which includes opioid abuse and dependence codes (61). While our study cohort excludes individuals with a recent prior OUD diagnosis or MOUD use, it is likely that many individuals classified as having “new” OUD had earlier histories of OUD diagnosis that could not be identified in our data. Additionally, although the quality of TAF data improved considerably by 2016 from its initial introduction in 2014, issues remain. The quality of race and ethnicity data was identified as “high concern” or “unusable” for over 20 states during the study period (38). Further, under-identification of OUD diagnoses and treatments is possible for managed care beneficiaries in some states due to incomplete managed care encounter data; however, we found results similar to the primary findings in sensitivity analyses excluding additional states based on potential data quality issues (38) and internal data quality assessments (Appendix A, Supplementary Methods, Table A1) (39). Our analysis was limited to medically-treated nonfatal overdoses, as we did not have access to mortality records or data on nonfatal overdoses that did not in medical treatment. While our findings reflect data from 2016 to 2019, prior to the onset of Covid-19 and shifts in the overdose epidemic, they illustrate critical relationships between patient and community characteristics, treatment receipt, and overdose, many of which are likely stable over time. Lastly, we used data from Medicaid enrollees meeting continuous enrollment and other inclusion criteria and may not generalize to other contexts.

Conclusions

Despite increasing from 2016–2019, MOUD remains underutilized among Medicaid enrollees newly diagnosed with OUD. Patients were at far lower risk of opioid overdose while receiving methadone, buprenorphine, and to a lesser extent naltrexone, compared to no receipt of MOUD. Results highlight the critical need for policy interventions to reduce variation in treatment receipt based on patient characteristics and where they live, and to increase linkage with MOUD and continuous receipt of treatment. Increasing access to methadone may be particularly important, given its large protective effect against overdose and the substantial policy and service delivery barriers to access.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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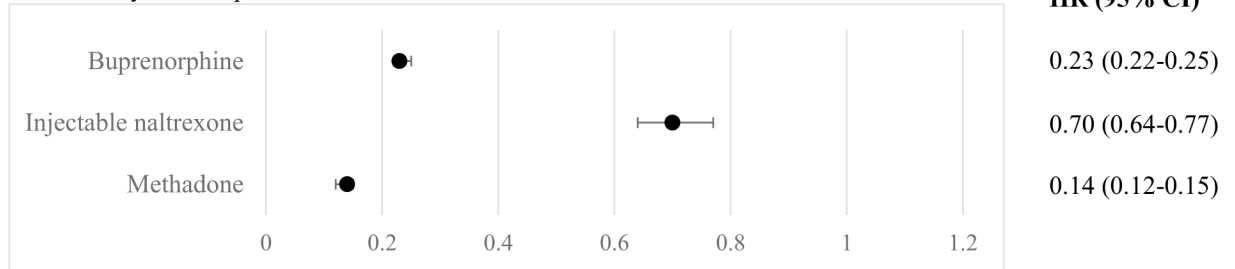
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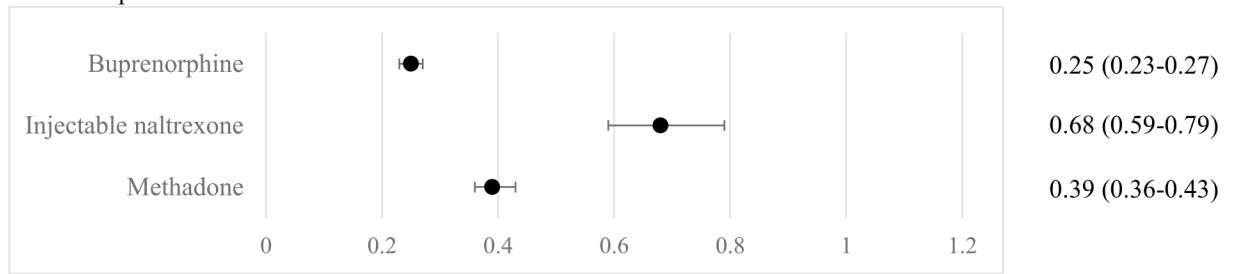
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A. Heroin/synthetic opioid overdose



B. Other opioid overdose

**Figure 1.**

Adjusted hazards of heroin/synthetic (A) and other opioid overdose (B) in 180 days after new OUD diagnosis, by MOUD type. Reference group is no MOUD. Estimates represent the hazard of overdose on days with compared to days without MOUD supply.

Abbreviations: OUD = opioid use disorder; MOUD = medication for opioid use disorder;

HR = hazard ratio; CI = confidence interval.

Table 1.
180-day MOUD receipt and medically-treated opioid overdose by individual, county, and state characteristics.

Characteristic	New OUD, No. (Col %)	MOUD Receipt, No. (Row %)	P value	Heroin/Synthetic Opioid Overdose, No. (Row %)	P value	Other Opioid Overdose, No. (Row %)	P value
No. (%) with outcome	1172200 (100)	361034 (30.8)		37641 (3.2)		21956 (1.9)	
Year			< .001		< .001		< .001
2016	266088 (22.7)	72804 (27.4)		9829 (3.7)		5291 (2.0)	
2017	342277 (29.2)	101392 (29.6)		12120 (3.5)		6725 (2.0)	
2018	299500 (25.6)	98107 (32.8)		9283 (3.1)		5574 (1.9)	
2019	264335 (22.6)	88731 (33.6)		6409 (2.4)		4366 (1.7)	
Race and Ethnicity			< .001		< .001		< .001
Black, non-Hispanic	167943 (14.3)	33473 (19.9)		4272 (2.5)		2914 (1.7)	
Hispanic	99295 (8.5)	28718 (28.9)		2967 (3.0)		1743 (1.8)	
Other, non-Hispanic	251314 (21.4)	72569 (28.9)		8180 (3.3)		5161 (2.1)	
White, non-Hispanic	653648 (55.8)	226274 (34.6)		22222 (3.4)		12138 (1.9)	
Age			< .001		< .001		< .001
18 to 29	309275 (26.4)	107150 (34.7)		13309 (4.3)		5909 (1.9)	
30 to 44	494832 (42.2)	176253 (35.6)		16218 (3.3)		8411 (1.7)	
45 to 64	368093 (31.4)	77631 (21.1)		8114 (2.2)		7636 (2.1)	
Sex			< .001		< .001		< .001
Female	607662 (51.8)	180785 (29.8)		14125 (2.3)		9556 (1.6)	
Male	564538 (48.2)	180249 (31.9)		23516 (4.2)		12400 (2.2)	
Comorbidities							
Alcohol use disorder	112266 (9.6)	24392 (21.7)	< .001	3383 (3.0)	< .001	2403 (2.1)	< .001
Cannabis use disorder	56139 (4.8)	9522 (17.0)	< .001	1437 (2.6)	< .001	938 (1.7)	< .001
Cocaine use disorder	37334 (3.2)	6746 (18.1)	< .001	1539 (4.1)	< .001	873 (2.3)	< .001
Sedative, hypnotic, or anxiolytic use disorder	11956 (1.0)	2676 (22.4)	< .001	427 (3.6)	.025	389 (3.3)	< .001
Stimulant use disorder (excluding cocaine)	57774 (4.9)	11553 (20.0)	< .001	1777 (3.1)	.058	941 (1.6)	< .001
Heroin/synthetic opioid poisoning	21548 (1.8)	8085 (37.5)	< .001	4780 (22.2)	< .001	1731 (8.0)	< .001
Other opioid poisoning	13265 (1.1)	3963 (29.9)	.020	1746 (13.2)	< .001	1926 (14.5)	< .001
Anxiety	263486 (22.5)	64901 (24.6)	< .001	6991 (2.7)	< .001	5807 (2.2)	< .001
Depression	205894 (17.6)	47377 (23.0)	< .001	5592 (2.7)	< .001	4563 (2.2)	< .001

Characteristic	New OUD, No. (Col %)	MOUD Receipt, No. (Row %)	P value	Heroin/Synthetic Opioid Overdose, No. (Row %)	P value	Other Opioid Overdose, No. (Row %)	P value
Chronic pain	577176 (49.2)	127319 (22.1)	< .001	13472 (2.3)	< .001	11420 (2.0)	< .001
Hepatitis C	59422 (5.1)	18065 (30.4)	.031	2975 (5.0)	< .001	1684 (2.8)	< .001
Prescriptions							
Baseline benzodiazepines	213801 (18.2)	48275 (22.6)	< .001	4874 (2.3)	< .001	5389 (2.5)	< .001
Baseline Opioids (non-MOUD)	468596 (40.0)	98804 (21.1)	< .001	8834 (1.9)	< .001	8929 (1.9)	.035
Follow-up Opioids (non-MOUD)	420948 (35.9)	64608 (15.4)	< .001	8014 (1.9)	< .001	8223 (2.0)	< .001
Any MOUD receipt during follow-up							
Buprenorphine	224131 (19.1)	224131 (100)	< .001	6823 (3.0)	< .001	3090 (1.4)	< .001
ER naltrexone	47819 (4.1)	47819 (100)	< .001	3068 (6.4)	< .001	1292 (2.7)	< .001
Methadone	105292 (9.0)	105292 (100)	< .001	2174 (2.1)	< .001	1375 (1.3)	< .001
County-level characteristics							
Median age, median (IQR)	38.1 (5.5)	37.9 (5.4)	< .001	38.3 (4.4)	.411	38.2 (5.3)	.270
Percentage white, non-Hispanic, median (IQR)	71.3 (34.3)	74.8 (32.4)	< .001	71.3 (29.1)	.004	70.1 (32.0)	< .001
SDI percentile, median (IQR)	54.0 (45.0)	52.0 (44.0)	< .001	49.0 (44.0)	< .001	52.0 (44.0)	< .001
Rurality			< .001		< .001		< .001
Non-metropolitan	227766 (19.4)	76512 (33.6)		3752 (1.7)		2848 (1.3)	
Metropolitan	944434 (80.6)	284522 (30.1)		33889 (3.6)		19108 (2.0)	
State-level characteristics							
Fentanyl penetration, median (IQR)	0.5 (0.9)	0.4 (0.8)	< .001	0.5 (0.9)	< .001	0.5 (0.9)	< .001
Medicaid expansion status	971282 (82.9)	315041 (32.4)	< .001	34339 (3.5)	< .001	18334 (1.9)	.011
Census region			< .001		< .001		< .001
Midwest	251763 (21.5)	77432 (30.8)		11182 (4.4)		5766 (2.3)	
South	388225 (33.1)	115038 (29.6)		8268 (2.1)		5742 (1.5)	
West	308468 (26.3)	89020 (28.9)		7175 (2.3)		4937 (1.6)	
Northeast	223744 (19.1)	79544 (35.6)		11016 (4.9)		5511 (2.5)	

Note: All outcomes are among Medicaid enrollees with new OUD. "Other" race and ethnicity includes American Indian and Alaska Native, Hawaiian/Pacific Islander, Multiracial, unspecified, and missing. Abbreviations: OUD = opioid use disorder; MOUD = medication for opioid use disorder; IQR = interquartile range; ER = extended release. P-values were derived from Pearson chi-square tests for categorical variables and t-tests for continuous variables.

Table 2. Adjusted hazards of MOUD receipt and medically-treated opioid overdose within 180 days of new OUD diagnosis ($N = 1,172,200$).

Characteristic	HR (95% CI)		
	MOUD receipt	Heroin/synthetic opioid overdose	Other opioid overdose
Year			
2016	REF	REF	REF
2017	1.03 (1.02, 1.04)	0.90 (0.87, 0.92)	0.98 (0.94, 1.01)
2018	1.14 (1.13, 1.15)	0.78 (0.76, 0.81)	0.94 (0.90, 0.97)
2019	1.21 (1.20, 1.22)	0.71 (0.68, 0.73)	0.94 (0.90, 0.99)
Race and Ethnicity			
White, non-Hispanic	REF	REF	REF
Black, non-Hispanic	0.68 (0.67, 0.69)	0.68 (0.66, 0.70)	0.80 (0.76, 0.83)
Hispanic	0.91 (0.90, 0.92)	0.89 (0.85, 0.92)	0.91 (0.86, 0.95)
Other, non-Hispanic	0.87 (0.86, 0.87)	0.94 (0.92, 0.97)	1.06 (1.02, 1.09)
Age			
18 to 29	1.06 (1.05, 1.07)	1.38 (1.34, 1.43)	1.13 (1.09, 1.18)
30 to 44	1.24 (1.23, 1.25)	1.19 (1.16, 1.23)	0.98 (0.95, 1.02)
55 to 64	REF	REF	REF
Sex			
Male	REF	REF	REF
Female	0.99 (0.98, 0.99)	0.65 (0.64, 0.66)	0.71 (0.69, 0.73)
Comorbidities			
Alcohol use disorder	0.78 (0.77, 0.79)	0.79 (0.76, 0.81)	0.91 (0.87, 0.95)
Cannabis use disorder	0.62 (0.61, 0.63)	0.67 (0.63, 0.70)	0.75 (0.70, 0.80)
Cocaine use disorder	0.75 (0.73, 0.77)	1.10 (1.04, 1.16)	0.96 (0.90, 1.03)
Sedative, hypnotic, or anxiolytic use disorder	0.93 (0.89, 0.96)	1.05 (0.95, 1.15)	1.24 (1.12, 1.37)
Stimulant use disorder (excluding cocaine)	0.62 (0.61, 0.63)	0.98 (0.93, 1.03)	0.85 (0.80, 0.91)
Heroin/synthetic opioid poisoning	1.06 (1.03, 1.08)	4.85 (4.69, 5.02)	2.31 (2.18, 2.44)
Other opioid poisoning	1.03 (1.00, 1.07)	1.73 (1.64, 1.82)	5.04 (4.77, 5.33)
Anxiety	0.96 (0.95, 0.97)	0.97 (0.94, 1.00)	1.05 (1.01, 1.08)
Depression	0.91 (0.90, 0.92)	0.96 (0.93, 0.99)	1.04 (1.00, 1.08)

Characteristic	HR (95% CI)		
	MOUD receipt	Heroin/synthetic opioid overdose	Other opioid overdose
Chronic Pain	0.79 (0.79, 0.80)	0.87 (0.85, 0.89)	0.98 (0.95, 1.01)
Hepatitis C	1.17 (1.15, 1.19)	1.60 (1.54, 1.66)	1.29 (1.23, 1.36)
Prescriptions			
Benzodiazepines	0.99 (0.98, 1.00)	1.01 (0.98, 1.05)	1.32 (1.28, 1.37)
Baseline opioids (non-MOUD)	0.90 (0.89, 0.91)	0.68 (0.66, 0.70)	0.83 (0.80, 0.86)
Follow-up opioids (non-MOUD)	0.34 (0.33, 0.35)	0.56 (0.53, 0.59)	1.39 (1.34, 1.45)
Daily receipt of MOUD during follow-up			
Buprenorphine	—	0.23 (0.22, 0.25)	0.25 (0.23, 0.27)
ER naltrexone	—	0.70 (0.64, 0.77)	0.68 (0.59, 0.79)
Methadone	—	0.14 (0.12, 0.15)	0.39 (0.36, 0.43)
County-level characteristics			
Median age (per 10 years)	0.99 (0.98, 1.00)	1.01 (0.97, 1.05)	1.11 (1.06, 1.17)
Percentage white, non-Hispanic (per 10% points)	1.03 (1.03, 1.04)	0.93 (0.92, 0.94)	0.93 (0.92, 0.94)
SDI percentile (per 10% points)	1.01 (1.01, 1.01)	0.97 (0.97, 0.98)	0.98 (0.97, 0.98)
Non-metropolitan	1.02 (1.01, 1.03)	0.56 (0.54, 0.58)	0.73 (0.70, 0.76)
State-level characteristics			
Fentanyl to heroin seizure ratio	1.11 (1.11, 1.12)	1.01 (0.99, 1.02)	1.05 (1.02, 1.07)
Medicaid expansion status	1.24 (1.23, 1.25)	1.58 (1.52, 1.64)	0.92 (0.88, 0.96)
Census region			
Northeast	REF	REF	REF
Midwest	0.91 (0.90, 0.92)	1.13 (1.10, 1.16)	1.04 (1.00, 1.08)
South	0.93 (0.92, 0.94)	0.63 (0.61, 0.65)	0.67 (0.64, 0.70)
West	0.98 (0.97, 0.99)	0.49 (0.47, 0.51)	0.66 (0.63, 0.70)

Note: Model additionally adjusts for baseline diagnosis of asthma, autism, bipolar disorder, cerebrovascular disease, chronic obstructive pulmonary disease, diabetes, heart failure, HIV, hypertension, intellectual disability, non-opioid drug poisoning, other mental health disorders, personality disorder, schizophrenia, sleep apnea, and baseline prescriptions for antidepressants and antipsychotics. Estimates for variables excluded from the table are shown in Table A5. Bold text denotes statistically significant estimates at $p < .05$. Where not specified, reference group is the absence of the characteristic. "Other" race and ethnicity includes American Indian and Alaska Native, Hawaiian/Pacific Islander, Multiracial, unspecified, and missing. Abbreviations: OUD = opioid use disorder; MOUD = medication for opioid use disorder; HR = hazard ratio; CI = confidence interval; ER = extended release; SDI = Social Deprivation Index.